



VAULTX CYBER TECH

BEYOND TODAY · BEFORE TOMORROW

SUMMER INTERNSHIP PROGRAM 2026

WEEK

01

AI FOUNDATIONS & ENVIRONMENT

PYTHON ENVIRONMENT / LLM FUNDAMENTALS / TOKENS & CONTEXT / API INTEGRATION / GIT WORKFLOW

6

TASKS

6

DELIVERABLES

BEGINNER

LEVEL

ONSITE / LAB

MODE

vaultxcybertech.com

info@vaultxcybertech.com · +92 306 2000928



WEEK FOCUS

Objective	Get a working development environment, understand what a language model actually is, and ship your first program that calls one.
Skills Targeted	Python environment LLM fundamentals Tokens & context API integration Git workflow
Tools & Stack	Python 3.11+ VS Code Git / GitHub venv Anthropic or OpenAI SDK python-dotenv

TASK LIST

01 Set up a professional dev environment

- Install Python 3.11 or newer, VS Code with the Python extension, and Git.
- Create a virtual environment and a **requirements.txt**. Never install packages globally.
- Create the GitHub repo **vaultx-ai-internship** with a README and a proper **.gitignore**.
- Make at least five meaningful commits this week. One giant commit at the end scores zero.

02 Build the vocabulary

- Write clear definitions in your own words for: token, context window, temperature, top-p, system prompt, embedding, hallucination, fine-tuning, inference.
- Explain the difference between a base model and an instruction-tuned model.
- Explain in three sentences why an LLM predicts text rather than "knows" facts.

03 Make your first API call

- Get an API key and store it in **.env**. Load it with python-dotenv.
- Write a script that sends a prompt and prints the response.
- Print the token usage from the response object and calculate the cost of that single call.

04 Experiment with generation parameters

- Run the same prompt at temperature 0, 0.7 and 1.0, three times each.
- Record the outputs in a table and describe how variance changes.
- State which temperature you would choose for: a support bot, a code generator, and a marketing copy tool. Justify each.

05 Build a reusable API wrapper

- Write a Python class or module that handles: sending messages, retry on failure, timeout, and token counting.
- Add proper error handling for rate limits and invalid keys. It must never crash on a bad response.
- You will reuse this module for the next seven weeks, so write it properly now.



06 Ship a working mini-tool

- Build a CLI tool that takes a block of text and returns a summary, key points and sentiment.
- It must accept input from a file or from a command-line argument.
- Record a demo video showing it running on at least two different inputs.

DELIVERABLES

#	DELIVERABLE	FORMAT
1	GitHub repo with venv, requirements.txt and .gitignore	REPO LINK
2	AI terminology glossary (9 terms, own words)	PDF / Markdown
3	First API call script with token & cost output	PY
4	Temperature comparison table with analysis	PDF
5	Reusable API wrapper module with error handling	PY
6	CLI summariser tool + demo video	PY + MP4

SUBMISSION PROTOCOL

Channel	Push code to your GitHub repo vaultx-ai-internship and submit the link on the VaultX LMS.
File Naming	VX_AI_W01_FirstName_LastName.pdf for the report; code in /week-01 of the repo.
Deadline	Sunday, 11:59 PM (PKT) of the assigned week.
Demo Video	2–4 minute screen recording showing the build actually running. Unrecorded work is not accepted.
Secrets Policy	No API keys in the repository. Use .env and commit a .env.example instead.



EVALUATION CRITERIA

CRITERION	WEIGHT	EVIDENCE REQUIRED
Environment set up correctly	20%	venv active, requirements pinned, secrets not committed
Conceptual understanding	20%	Glossary is in own words, not copy-pasted definitions
Working API integration	25%	Demo video shows the tool running end to end
Code quality of wrapper	20%	Handles errors, retries, and does not crash on bad input
Git discipline	15%	Five or more commits with descriptive messages